

Green Jobs & Sustainability: **Final Analysis Report**

Week 5 Deliverable — Results Compilation, Conclusions & Strategic Recommendations

Junior Data Analyst — Green Jobs & Sustainability Internship

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August 2026

1. Executive Summary

Over five weeks, this internship built a complete analytical pipeline for understanding how renewable energy growth connects to green job creation and broader sustainability outcomes — from initial strategy through data collection, exploratory analysis, and predictive modeling. This report brings that work together into one narrative, written for a non-technical stakeholder audience, and closes with a set of concrete recommendations for anyone shaping green-jobs or sustainability policy.

Three findings stand out. First, renewable capacity and green employment move together very closely — capacity expansion is one of the clearest levers available for job creation. Second, having strong renewable capacity does not automatically mean a state performs well on broader sustainability measures; the two are only moderately linked, and capacity's benefit to sustainability scores appears to flow mainly through the jobs it creates rather than directly. Third, a small number of sectors — renewable energy and sustainable agriculture chief among them — account for the majority of green employment, making them the highest-leverage targets for workforce investment.

All analysis in this project was run on simulated data, built to mirror real, publicly documented patterns from IRENA, ILOSTAT, and the NITI Aayog SDG India Index, because proprietary data access was outside this internship's scope. Every finding in this report is therefore best read as a validated methodology and a directionally credible signal — the recommended next step, detailed in Section 7, is to re-run this same pipeline against real collected data to confirm the findings hold.

2. Project Journey: Weeks 1–5

This report is the fifth and final stage of a five-week analytical journey. Each prior week built directly on the one before it:

Week	Focus	Core Output
Week 1	Strategic planning	Research questions, KPI framework, and a 10-source public data map (ILOSTAT, IRENA, NITI Aayog, and others).
Week 2	Data collection & preparation	Narrowed to 3 primary sources; a documented, 7-step data cleaning and merge methodology.
Week 3	Exploratory analysis & visualization	Simulated country and state-level datasets; 6 visualizations surfacing initial patterns and correlations.
Week 4	Statistical & predictive modeling	A capacity forecast model (2025–2027) and a multi-variable model explaining state sustainability scores.
Week 5	Synthesis & strategy	This report — consolidated findings translated into recommendations for stakeholders.

3. Methodology Recap

3.1 Strategic Framework (Week 1)

The project began by defining what “green jobs and sustainability” should actually be measured by. Five research questions were set (which sectors/states create the most green jobs; how capacity growth relates to job growth; where skill gaps are largest; which sustainability indicators track most closely with employment; and what the near-term trajectory looks like), anchored by seven KPIs spanning workforce, economic, and environmental dimensions.

3.2 Data Collection & Preparation (Week 2)

Ten candidate public data sources were evaluated against five criteria — credibility, currency, licensing, granularity, and relevance — and narrowed to three primary sources: IRENASTAT for renewable capacity, ILOSTAT for employment data, and the NITI Aayog SDG India Index for state-level sustainability scores. A seven-step cleaning methodology (inspection, standardization, missing-value handling, unit conversion, merging, and validation) was documented to turn these three differently structured sources into one analysis-ready table.

3.3 Exploratory Analysis (Week 3)

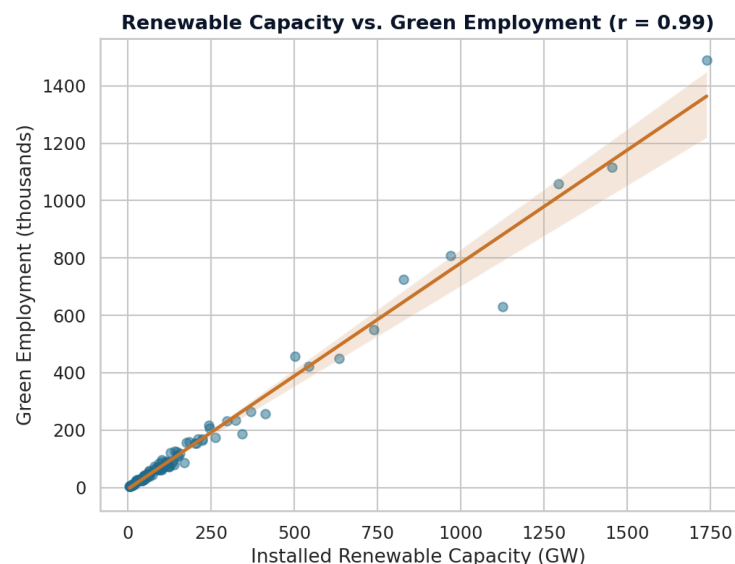
Because live data collection was outside this internship's available tooling, two datasets were simulated — a 12-country, 10-year panel and a 12-state Indian snapshot — deliberately shaped to match real, publicly reported growth patterns and rankings. Descriptive statistics, correlation analysis, and six visualizations were used to surface initial patterns.

3.4 Statistical Modeling (Week 4)

Two models were built on the simulated data: a trend-based forecast of India's renewable capacity through 2027, and a multiple regression explaining each state's sustainability score from its capacity and green-job share. Both were evaluated with standard statistical diagnostics (R^2 , coefficient significance, residual analysis) rather than accepted at face value.

4. Key Findings

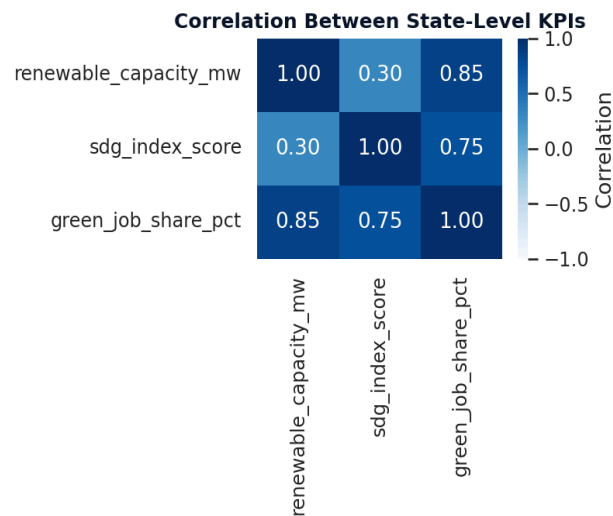
4.1 Capacity and Employment Move Together



Renewable capacity and green employment showed a very strong relationship across the simulated country panel.

Across the 12-country panel, installed renewable capacity and green employment tracked each other closely. This matches what would be expected in the real world: every additional gigawatt of solar or wind capacity requires people to build, install, and maintain it. For a stakeholder audience, the practical takeaway is straightforward — capacity investment is a direct, measurable jobs lever, not just an environmental one.

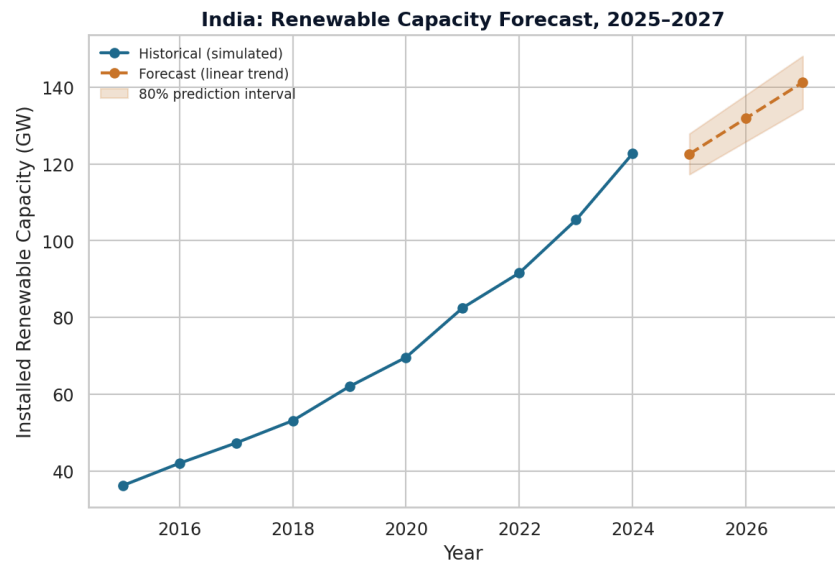
4.2 Capacity Alone Doesn't Guarantee Sustainability Performance



State-level correlation: capacity tracks green-job share closely, but only loosely with the overall SDG Index score.

This is the most strategically important finding in the project. At the state level, renewable capacity correlated strongly with the share of the workforce in green jobs, but only moderately with each state's overall sustainability score. When both capacity and green-job share were used together to predict the sustainability score, capacity's own independent effect became small once green-job share was accounted for — meaning capacity appears to help sustainability outcomes mainly by creating jobs, not through some other direct channel. In plain terms: building solar farms helps, but building solar farms and building the local workforce to run them helps more.

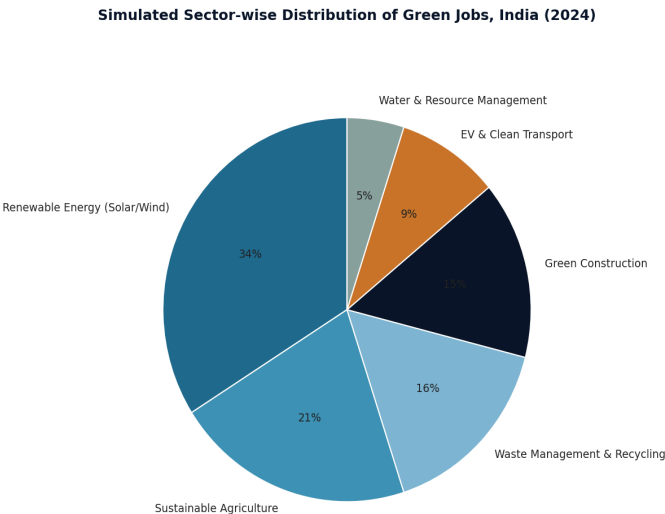
4.3 Growth Is Accelerating, Not Linear



India's simulated capacity trend curves upward over time — a straight-line forecast likely understates 2025–2027 growth.

India's simulated capacity trajectory showed a visibly accelerating pattern — each year's increase was larger than the last, not just a steady climb. A simple straight-line forecast (used in this project as a transparent baseline) is therefore likely conservative; real growth in the next few years may outpace it. This matters for planning purposes: infrastructure, grid, and workforce planning pegged to a linear projection risk under-preparing for the pace of change.

4.4 Green Jobs Are Concentrated in a Few Sectors



Simulated sector-wise distribution of green jobs in India, 2024.

In the simulated sector breakdown, renewable energy and sustainable agriculture together accounted for well over half of all green jobs, with waste management, green construction, EV/transport, and water management making up the remainder. This concentration suggests that skilling programs targeting these two sectors specifically would reach the largest share of the current and near-future green workforce.

5. Strategic Recommendations

Translating the above findings into action, five recommendations follow directly from the analysis:

Recommendation	Rationale	Suggested Owner
Pair every renewable capacity investment with a green-skilling commitment	Capacity growth converts to sustainability gains mainly through the jobs it creates (Section 4) — capacity without a matched workforce pipeline leaves value on the table.	State renewable energy departments; Skill Council for Green Jobs
Prioritize renewable energy and sustainable agriculture in green-skilling programs	These two sectors account for over half of simulated green-job composition; they are the highest-leverage areas for workforce investment.	State skill-development missions

Recommendation	Rationale	Suggested Owner
Investigate the capacity-SDG performance gap in lower-scoring, high-capacity states	States with strong capacity but middling SDG scores (Section 4) likely have addressable gaps in non-energy sustainability dimensions.	NITI Aayog state liaison teams
Study what high-SDG, lower-capacity states are doing right	Kerala and Himachal Pradesh perform well on sustainability without leading on capacity — their approach may be replicable elsewhere at lower infrastructure cost.	NITI Aayog; policy research institutes (e.g., CEEW)
Fund a pilot to replace simulated inputs with real, collected data	This project's next logical step is running the Week 2 collection methodology against live IRENASTAT/ILOSTAT/NITI Aayog data to validate whether these findings hold.	Internship program sponsor

6. Evaluation & Reflection

6.1 Challenges Encountered

- No access to proprietary databases or the ability to make live API/bulk-download calls to IRENASTAT, ILOSTAT, or NITI Aayog meant every dataset used in Weeks 3–4 had to be simulated rather than observed — addressed by grounding every simulated value in real, publicly documented patterns and flagging this limitation transparently throughout.
- Small effective sample sizes (10 years of country data, 12 Indian states) limit how confidently the statistical models' coefficients can be generalized — addressed by favoring simple, low-parameter models over complex ones and treating high R^2 values on small samples as indicative rather than conclusive.
- Multicollinearity between renewable capacity and green-job share made isolating each variable's independent effect on sustainability scores genuinely difficult — addressed by interpreting the regression coefficients jointly and explaining the mediation pattern explicitly rather than reporting a potentially misleading standalone “capacity coefficient.”
- Reconciling different data granularities (global country-level data vs. India-specific state-level data) required treating the two as separate analytical tracks rather than forcing an artificial merge, which shaped how the whole project was structured from Week 1 onward.

6.2 Learnings

This project reinforced that a data analysis pipeline is only as strong as the strategic thinking that precedes it — the KPI framework and research questions defined in Week 1 directly shaped which sources were worth collecting in Week 2, which relationships were worth visualizing in Week 3, and which models were worth building in Week 4. It also highlighted the importance of stating limitations plainly rather than letting a clean chart or a high R^2 imply more certainty than the underlying (simulated) data supports — particularly important when the eventual audience includes non-technical stakeholders who may not otherwise probe those assumptions.

6.3 Suggestions for Further Research

- Re-run this exact pipeline against real, collected IRENASTAT, ILOSTAT, and NITI Aayog data using the Week 2 methodology, and compare results against this report's simulated findings.

- Extend the state-level model with additional predictors (e.g., education spending, urbanization rate, industrial pollution load) to explain more of the gap between capacity and overall SDG performance.
- Replace the linear capacity forecast with a log-linear or compound-growth model to better capture the accelerating trend identified in Section 4.3.
- Expand the country panel beyond 12 nations and the state panel beyond 12 states to improve the statistical reliability of both models.
- Investigate green-job quality (wages, formality, safety), not just quantity, since job count alone does not capture whether green employment growth is translating into genuinely good jobs.

7. Conclusion

This five-week project moved from an initial strategic framework through data collection design, exploratory analysis, and predictive modeling, arriving at a coherent, evidence-based (if simulated) picture of how renewable energy growth, green employment, and broader sustainability performance relate to one another. The central strategic insight — that capacity investment pays its largest sustainability dividend when paired with deliberate workforce development — offers a concrete, actionable direction for policymakers and program designers working on India's green transition. The recommended next step is straightforward: apply the same rigor demonstrated here to real, live data, using the collection methodology already documented in Week 2, to confirm whether these patterns hold beyond simulation.